Chapter 2
Basic Structures Sets, Functions Sequences, and Sums

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Objectives

- Sets
- Set operations
- Functions
- Sequences
- Summations

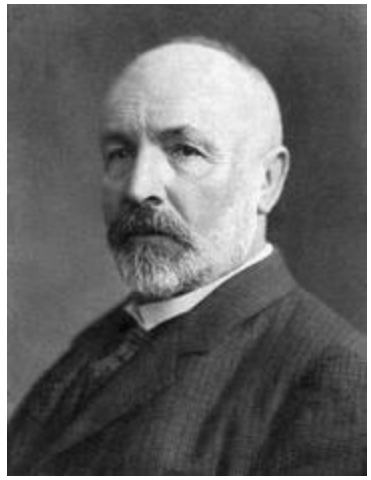
2.1- Sets

- An unordered collection of objects
- The objects in a set are called the elements, or members. A set is said to contain its elements.
- We write $a \in A$ to denote that a an element of the set A. The notation $a \notin A$ denote that a is not an element of the set A.
- Example:

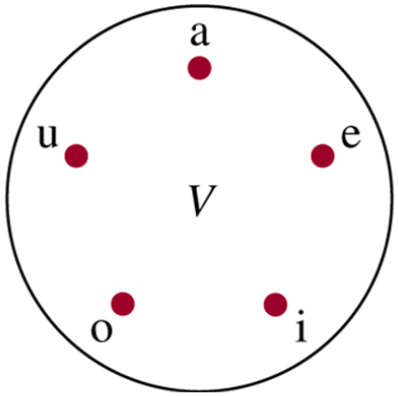
  O = { 1, 3, 5, 7, 9}

$$O = \{x \in \not\subset^{+} \mid x \text{ is odd and } x < 10\}$$

$$V = \{a, u, o, i, e\}$$



G. Cantor



Venn Diagram for V

MEMBERS/ELEMENTS of a set

- ♥ ∈ {1, a, ♥} // ♥ is in {1, a, ♥}
- ♥ ∉ {♦, ≈, {♥}, f} // ♥ is not an element of this set
- a ∉ {{a}, b} {a} ∉ {a, b}
- {a} ∈ {{a}, b} // {a} is a member of {{a}, b}

Definitions:

Finite set: Set has n elements, n is a nonnegative integer

A set is an ***infinite*** set if it is not finite

WAYS TO DEFINE SETS

- **Method 1.** Roster method. List all members:

V = {a, i, o, e, u}

Z = {…, -2, -1, 0, 1, 2, …}

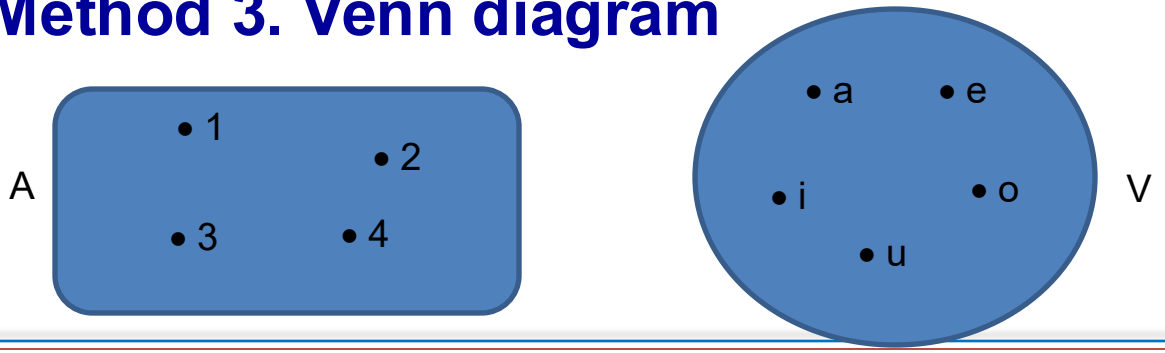
- **Method 2. Set builder notation.** Describe a set by **properties**

A = {x | P(x)}

A = {x | x is a positive integer and x < 5}

Read: A is the set of positive integers such that less than 5.

- **Method 3. Venn diagram**



NULL SET. EMPTY SET

- **Null set/empty set**: set has no element.

A = {x | x is a month with 32 days}

A = { } // empty set

- Symbo**NOTE:** l: { } or ∅
- ∅ belongs to {∅}.

So, {∅} is different from ∅

$$\emptyset \neq \{\emptyset\}$$

(Way to remember: empty folder ≠ folder containing empty folder)

SOME IMPORTANT SETS

- N: the set of natural numbers

  $N = \{0, 1, 2, 3, \ldots\}$

- Z: the set of integers

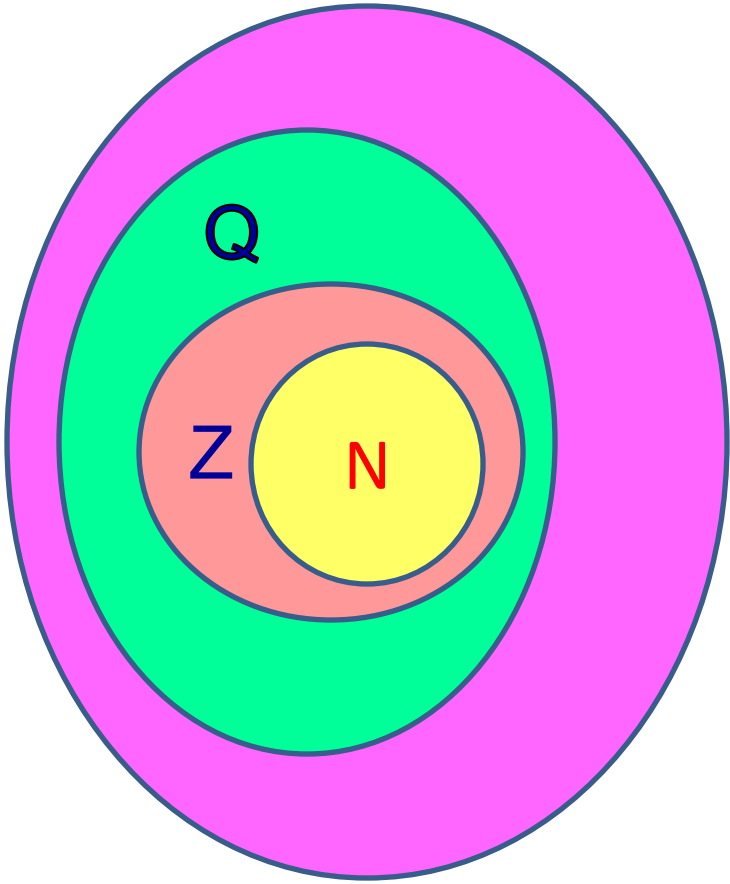
  $Z = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$

- Q = the set of rational numbers

  $Q = \{\frac{p}{q} \mid p, q \in Z, q \neq 0\}$

$Q = \{\ldots, -1/3, 0, 2, -3, 5/7, \ldots\}$

- R: the set of real numbers



CARDINALITY OF A SET

- If a set has n elements, we say its cardinality is n.
- A = {a, b, c} ➔ cardinality of A, |A| = 3
- B = ∅ ➔ |B| = 0
- C = {∅} ➔ |C| = 1
- D = {a, {a}} ➔ |D| = 2
- E = {a, a, a, a}
- ➔ |E| = 1
- If V is the set of all vowels in English, then |V| = 5
- |N| = ∞

SUBSETS

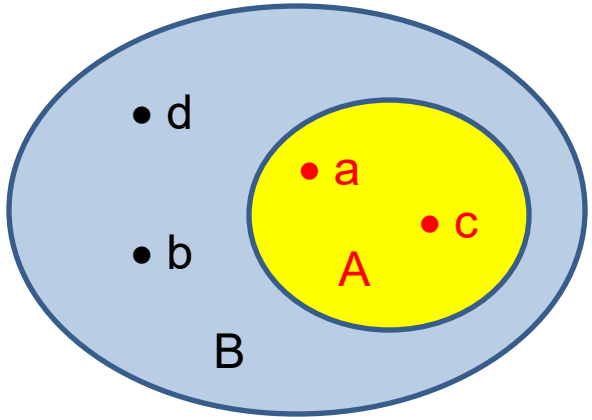
- If every element of A is in B, then we say A is a **SUBSET** of B.

$$\forall x(x \in A \rightarrow x \in B)$$

- Symbol: $A \subseteq B$

  $A \subset B$ (if $A \neq B$)

- Venn diagram:



Venn diagram for $A \subseteq B$

- Examples:

$\{a, c\} \subset \{a, b, c, d\}$

$N \subset Z$

$Z \subset Q$

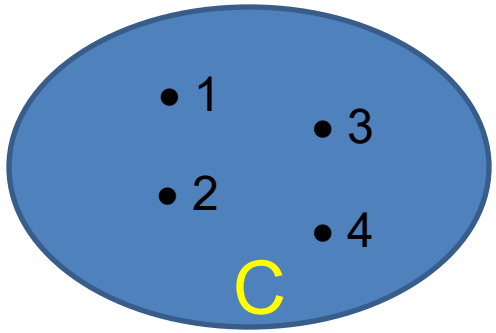
$Q \subset R$

HOW TO SHOW THAT A ⊆ B and A□B

- To show A ⊆ B: show that if x∈A, then x∈B
- To show A□B: find only one x∈A such that x∉B

EQUALITY

- Two sets A and B are **equal** if they have the same elements no matter how they are defined.
- Symbol: A = B
- A = {x | x is a positive integer less than 5}
- B = {1, 2, 3, 4, 3, 3}
- D = {3, 2, 4, 1}
- A = B = C = D



EQUALITY

- To show that A = B:
- **Method 1.** Show that $A \subseteq B$ and $B \subseteq A$
- **Method 2.** Use set builder notation.
- **Method 3.** Use membership table.

Theorem 1

For every set S ,

i) $\varnothing \subseteq S$ ii) $S \subseteq S$

Proof

i) $(x \in \varnothing) \equiv \text{False}$

So $\forall x\left(x \in \varnothing \rightarrow x \in S\right) \equiv \text{True}$

ii) $\forall x\left(x \in S \rightarrow x \in S\right) \equiv \text{True}$

Power Sets

Given a set S, **power set** P(S) of S is a set of all subsets of the set S.

S= {1,2,3}

P(S)= {Ø, {1}, {2},{3}, {1,2}, {1,3},{2,3}, {1,2,3}}

If A has n elements, then P(A) has 2^n elements.
($|A| = n \rightarrow |P(A)| = 2^n$)

Cartesian Products

- The **ordered** n-tuple $(a_1,a_2,\ldots,a_n)$ is the **ordered collection** that has a_1 as its first element, a_2 as its second element, …, and a_n as its nth element.
- Let A and B be sets. The Cartesian product of A and B, denoted by AxB,

$$A \times B = \{(a,b) | a \in A, b \in B\}$$

For example

$A=\{a,b\}$ $B=\{1,2,3\}$

$$A \times B = \{(a,1),(a,2),(a,3),(b,1),(b,2),(b,3)\}$$

Cartesian Products…

- The Cartesian product of $A_1,A_2,\ldots,A_n$, denoted $A_1xA_2x\ldots xA_n$, is the set of ordered n-tuples $(a_1,a_2,\ldots,a_n)$,

$$A_1 \times A_2 \times \ldots \times A_n = \left\{ (a_1, a_2, \ldots, a_n) \middle| a_i \in A_i, \forall i = \overline{1,n} \right\}$$

For example

$A=\{a,b\}$ $B=\{1,2,3\}, C=\{0,1\}$

- AxBxC= {(a,1,0),(a,1,1),(a,2,0),(a,2,1),(a,3,0),(a,3,1),
(b,1,0),(b,1,1),(b,2,0),(b,2,1),(b,3,0),(b,3,1) }

1. Determine whether each of these statements is true or false.

a) $0 \in \emptyset$ **b)** $\emptyset \in \{0\}$ **c)** $\{0\} \subset \emptyset$ **d)** $\emptyset \subset \{0\}$

e) $\{0\} \in \{0\}$ **f)** $\{0\} \subset \{0\}$ **g)** $\{\emptyset\} \subseteq \{\emptyset\}$

1. Determine whether each of these statements is true or false.

a) $0 \in \emptyset$ **b)** $\emptyset \in \{0\}$ **c)** $\{0\} \subset \emptyset$ **d)** $\emptyset \subset \{0\}$

e) $\{0\} \in \{0\}$ **f)** $\{0\} \subset \{0\}$ **g)** $\{\emptyset\} \subseteq \{\emptyset\}$

a) False **b)** False **c)** False **d)** True

e) False **f)** False **g)** True

2. Determine whether these statements are true or false.

a) $\emptyset \in \{\emptyset\}$ **b)** $\emptyset \in \{\emptyset, \{\emptyset\}\}$ **c)** $\{\emptyset\} \in \{\emptyset\}$ **d)** $\{\emptyset\} \in \{\{\emptyset\}\}$

e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$ **f)** $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$ **g)** $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

2. Determine whether these statements are true or false.

a) $\emptyset \in \{\emptyset\}$ **b)** $\emptyset \in \{\emptyset, \{\emptyset\}\}$ **c)** $\{\emptyset\} \in \{\emptyset\}$ **d)** $\{\emptyset\} \in \{\{\emptyset\}\}$

e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$ **f)** $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$ **g)** $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

a) True **b)** True **c)** False **d)** True

e) True **f)** True **g)** False

3. Determine whether each of these statements is true or false.

a) $x \in \{x\}$ **b)** $\{x\} \subseteq \{x\}$ **c)** $\{x\} \in \{x\}$

d) $\{x\} \in \{\{x\}\}$ **e)** $\emptyset \subseteq \{x\}$ **f)** $\emptyset \in \{x\}$

a) True **b)** True **c)** False

d) True **e)** True **f)** False

1. List the members of these sets.

a) $\{x \mid x$ is a real number such that $x^2 = 1\}$

b) $\{x \mid x$ is a positive integer less than 12$\}$

c) $\{x \mid x$ is the square of an integer and $x < 100\}$

d) $\{x \mid x$ is an integer such that $x^2 = 2\}$

2. Use set builder notation to give a description of each of these sets.

a) {0, 3, 6, 9, 12}

b) {-3, -2, -1, 0, 1, 2, 3}

c) {*m*, *n*, *o*, *p*}

1. List the members of these sets.

a) {x | x is a real number such that $x^2 = 1$}

b) {x | x is a positive integer less than 12}

c) {x | x is the square of an integer and $x < 100$}

d) {x | x is an integer such that $x^2 = 2$}

2. Use set builder notation to give a description of each of these sets.

a) {0, 3, 6, 9, 12}

b) {-3, -2, -1, 0, 1, 2, 3}

c) {*m*, *n*, *o*, *p*}

(a) $\{x \in \mathbb{N} | x \text{ is a multiple of 3 and } x \leq 12\}$

(b) $\{x \in \mathbb{Z} | -3 \leq x \leq 3\}$

(c) $\{x | x \text{ is a letter in the alphabet from } m \text{ to } p\}$

19. What is the cardinality of each of these sets?

a) $\{a\}$ **b)** $\{\{a\}\}$

c) $\{a, \{a\}\}$ **d)** $\{a, \{a\}, \{a, \{a\}\}\}$

20. What is the cardinality of each of these sets?

a) $\emptyset$ **b)** $\{\emptyset\}$

c) $\{\emptyset, \{\emptyset\}\}$ **d)** $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$

21. Find the power set of each of these sets, where a and b are distinct elements.

a) $\{a\}$ **b)** $\{a, b\}$ **c)** $\{\emptyset, \{\emptyset\}\}$

23. How many elements does each of these sets have where a and b are distinct elements?

a) $P(\{a, b, \{a, b\}\})$

b) $P(\{\emptyset, a, \{a\}, \{\{a\}\}\})$

c) $P(P(\emptyset))$

24. Determine whether each of these sets is the power set of a set, where a and b are distinct elements.

a) $\emptyset$

b) $\{\emptyset, \{a\}\}$

c) $\{\emptyset, \{a\}, \{\emptyset, a\}\}$

d) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

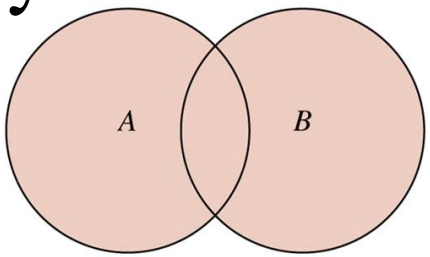
2.2 - SET OPERATIONS.

Set operations	Symbols
Union	$A \cup B$
Intersection	$A \cap B$
Difference	$A - B$
Symmetric difference	$A \oplus B$
Complement	$\bar{A}$

2.2- Set Operations

- The *Union* of sets A and B, denoted by $A \cup B$

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

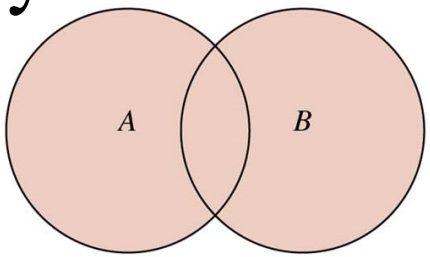


A	B	A ∪ B
{a, b, c}	{c, d}	
{a, b, c}	{a, b, c, d, e}	
∅	{1, 2, 3}	
The set of all students who are from Long An	The set of all students who are from Phu Yen	

2.2- Set Operations

- The *Union* of sets A and B, denoted by $A \cup B$

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



A	B	A ∪ B
{a, b, c}	{c, d}	{a, b, c, d}
{a, b, c}	{a, b, c, d, e}	{a, b, c, d, e}
∅	{1, 2, 3}	{1, 2, 3}
The set of all students who are from Long An	The set of all students who are from Phu Yen	The set of all students who are from Long An OR Phu Yen

MEMBERSHIP TABLE FOR A ∪ B

0: NOT A MEMBER OF THE SET
1: MEMBER OF THE SET

A	B	A ∪ B
0	0	0
1	0	1
0	1	1
1	1	1

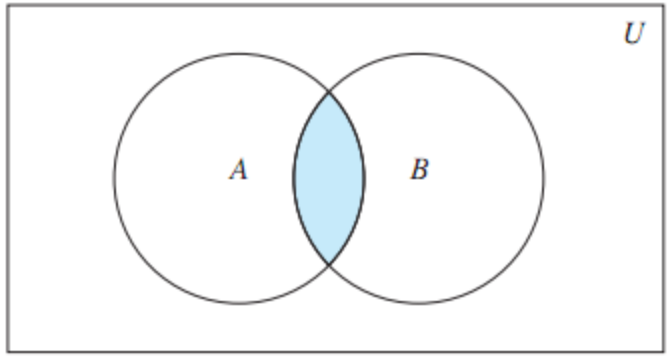
Consider x

INTERSECTION

- Symbol: $A \cap B$
- Set builder:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

- Examples.



$A \cap B$ is shaded.

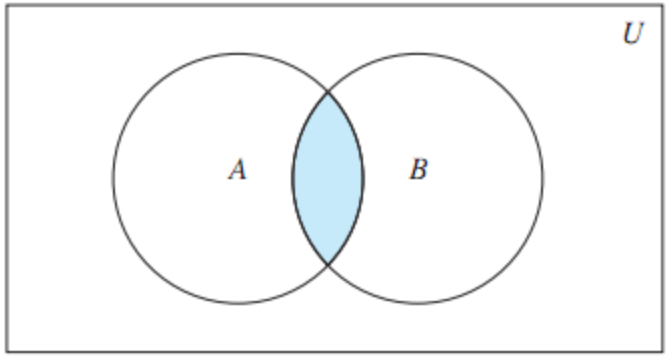
A	B	$A \cap B$
{a, b, c}	{c, d, e}	
{a, b, c}	{a, b, c, d, e}	
{1, 2, 3}	{4, 5}	
Set of all bit strings of length 4 start with 1	Set of all bit strings end with 00	

INTERSECTION

- Symbol: $A \cap B$
- Set builder:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

- Examples.



$A \cap B$ is shaded.

A	B	A ∩ B
{a, b, c}	{c, d, e}	{c}
{a, b, c}	{a, b, c, d, e}	{a, b, c}
{1, 2, 3}	{4, 5}	∅ // **disjoint** sets
Set of all bit strings of length 4 start with 1	Set of all bit strings end with 00	Set of all bit strings of length 4 start with 1**AND** ends 00

MEMBERSHIP TABLE FOR A ∩ B

0: NOT A MEMBER OF THE SET
1: MEMBER OF THE SET

A	B	A ∩ B
0	0	?
1	0	?
0	1	?
1	1	?

CARDINALITY OF A∪B

$|A \cup B| = |A| + |B|$??? NOT TRUE

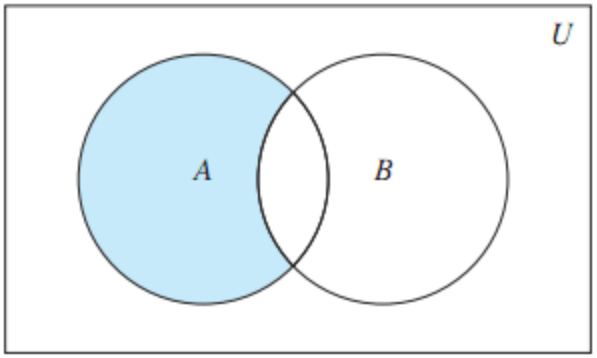
$|A \cup B| = |A| + |B| - |A \cap B|$

- For example
- A = {1, 2, 3, 4, 5} ➔ |A| = 5
- B = {4, 5, 6, 7} ➔ |B| = 4
- A ∪ B = {1, 2, 3, 4, 5, 6, 7} ➔ |A∪B| = 7
- A∩B = {4, 5} ➔ |A∩B| = 2

DIFFERENCE

- Symbol: A – B // difference of A and B
- Set builder:

$A - B = \{x \mid x \in A \wedge x \notin \mathbf{B}\}$



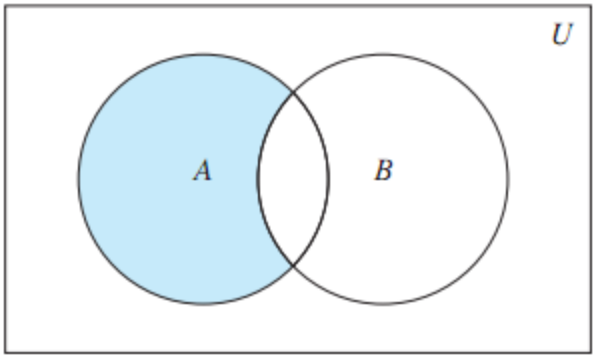
$A - B$ is shaded.

A	B	A – B
{a, b, c}	{c, d, e}	
{a, b, c}	{a, b, c, d, e}	
{1, 2, 3}	{4, 5}	
The set of all bit strings of length 4 begin with 1	The set of all bit strings of length 4 end with 00	

DIFFERENCE

- Symbol: A – B // difference of A and B
- Set builder:

$$A - B = \{x \mid x \in A \wedge x \notin \mathbf{B}\}$$



$A - B$ is shaded.

A	B	A – B
{a, b, c}	{c, d, e}	{a, b}
{a, b, c}	{a, b, c, d, e}	{ }
{1, 2, 3}	{4, 5}	{1, 2, 3}
The set of all bit strings of length 4 begin with 1	The set of all bit strings of length 4 end with 00	The set of all bit string of length 4 begin with 1 AND NOT end with 00

MEMBERSHIP TABLE FOR A – B

0: NOT A MEMBER OF THE SET
1: MEMBER OF THE SET

A	B	A – B
0	0	0
1	0	1
0	1	0
1	1	0

COMPLEMENT OF A SET

- UNIVERSAL SET **U**: set contains all objects under consideration.
- Complement of a set A, denoted by $\bar{A}$ is the defined by:

$\bar{A}$ = U – A

$$x \in A \leftrightarrow x \notin \bar{A}$$

Example 1: U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

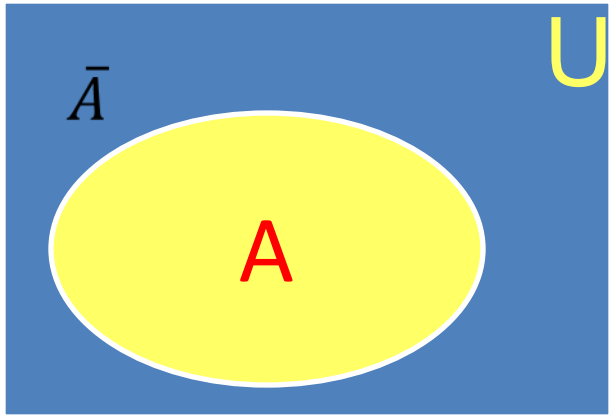
A = {1, 3, 5} ➔ $\bar{A}$ = {2, 4, 6, 7, 8, 9, 10}

Example 2: U is the set of all letters of the English alphabet.

A = {e, a, l, o, u}

➔ $\bar{A}$ = {b, c, d, f, g, h, j, k, l, m, n ,p, q, r, s, t, v, w, x, y, z}.

VENN DIAGRAM. MEMBERSHIP TABLE OF $\bar{A}$



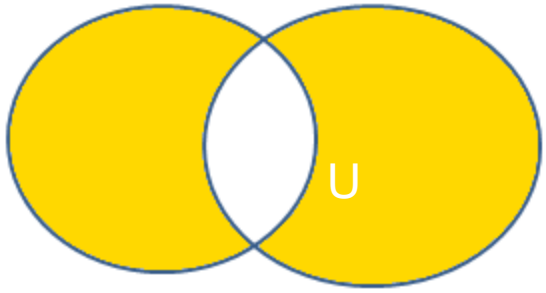
A	$\bar{A}$
0	1
1	0

SYMMETRIC DIFFERENCE

- The **symmetric difference** of A and B is the set:

$$(A - B) \cup (B - A)$$

- Symbol: **A⊕B**



A ⊕ B is colored

MEMBERSHIP TABLE

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

SET IDENTITIES

Identity		Name
$A \cap U = A$	$A \cup \varnothing = A$	Identity laws
$A \cup U = U$	$A \cap \varnothing = \varnothing$	Domination laws
$A \cup A = A$	$A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$		Complementation law
$A \cup B = B \cup A$	$A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$	$A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$	$\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$	$A \cap \overline{A} = \varnothing$	Complement laws

PROOFS OF $\overline{A \cap B} = \overline{A} \cup \overline{B}$

MEMBERSHIP TABLES

A	B	$\overline{A}$	$\overline{B}$	$\overline{A} \cup \overline{B}$	$A \cap B$	$\overline{A \cap B}$
0	0	1	1	1	0	1
0	1	1	0	1	0	1
1	0	0	1	1	0	1
1	1	0	0	0	1	0

REPRESENTING SETS IN COMPUTER

Ex1. U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} // i = 1..10

A = {1, 3, 5}, B = {1, 2, 4, 5, 7, 9}

➔ A = "1 0 1 0 1 0 0 0 0 0" // $[a_i]$: $a_i = 1$ iff $i \in A$

➔ B = "1 1 0 1 1 0 1 0 1 0" // $[b_i]$: $b_i = 1$ iff $i \in B$

HOW TO FIND A∩B? ← bitwise AND

A∩B = "1 0 0 0 1 0 0 0 0 0" // $[c_i] := a_i \wedge b_i$

➔ A∩B = {1, 5} // If $[c_i] = 1$ then print i in A∩B

Example 2.

U = {a, b, c, d, e}

A = {b, d} ➔ A = "0 1 0 1 0"

Generalized Unions and Intersections

$$A_1 \cap A_2 \cap A_3 \cap \ldots \cap A_n = \bigcap_{i=1}^{n} A_i = \{x \mid x \in A_i, \forall i = 1, 2, \ldots, n\}$$

$$A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_n = \bigcup_{i=1}^{n} A_i$$

$$= \{x \mid x \in A_1 \vee x \in A_2 \vee x \in A_3 \vee \ldots \vee x \in A_n\}$$

Computer Representation of Sets

- Use bit string U={1,2,3,4,5,6,7,8,9,10}
- A= {1,3,5,7,9 } → A = "1010101010"
- B= { 1,8,9} → B = "1000000110"

Computer Representation of Sets

- A = “1010101010”
- B = “1000000110”

$$A \cup B = 10\ 1010\ 1010 \vee 10\ 0000\ 0110 = 10\ 1010\ 1110$$

$$A \cup B = \{1, 3, 5, 7, 8, 9\}$$

$$A \cap B = 10\ 1010\ 1010 \wedge 10\ 0000\ 0110 = 10\ 0000\ 0010$$

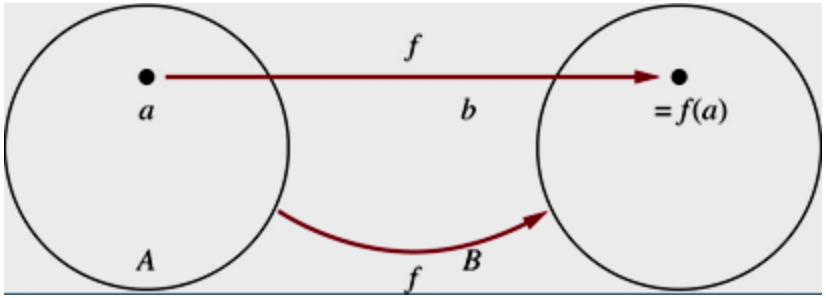
$$A \cap B = \{1, 9\}$$

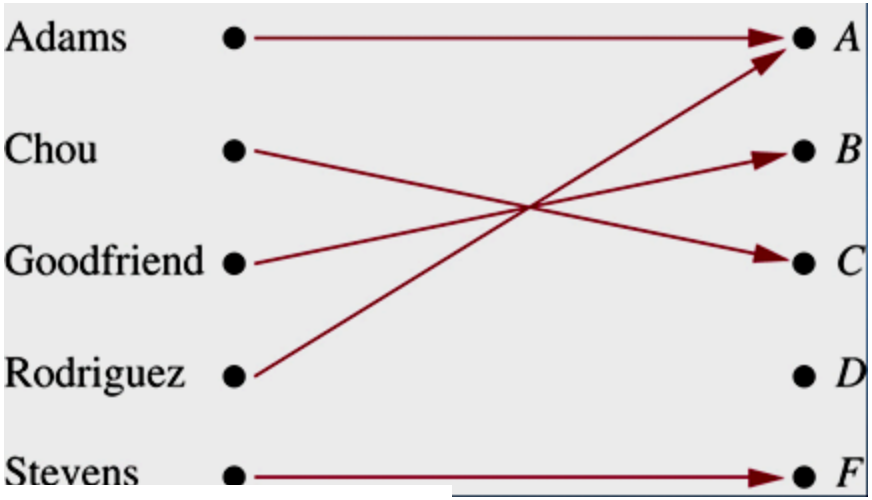
2.3- Functions/Mapping/ Transformation - Ánh Xạ

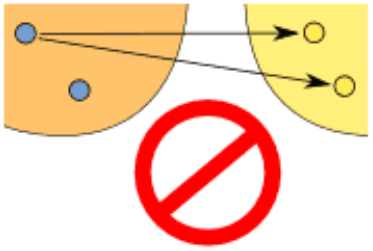
- Introduction
- Some important Functions
- One-to-One and Onto Functions
- Inverse Function
- Composition Functions

2.3- Functions/Mapping/ Transformation - Ánh Xạ

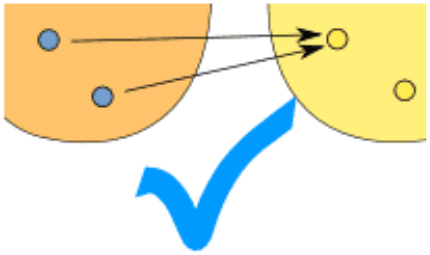
- **f: A → B** function f from A to B (or function f maps A to B)
- A: **domain** of f
- B: **codomain** of f







(one-to-many)
This is **NOT** OK in a function

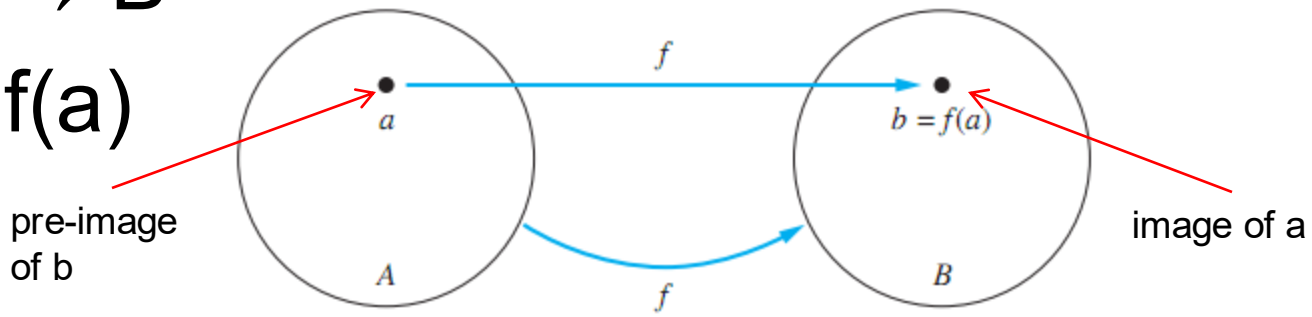


(many-to-one)
But this **is** OK in a function

2.3- Functions/Mapping/ Transformation - Ánh Xạ

$f: A \rightarrow B$

$b = f(a)$



Three parts of a function.

$f: A \rightarrow B$

❶ A: domain

❷ B: codomain

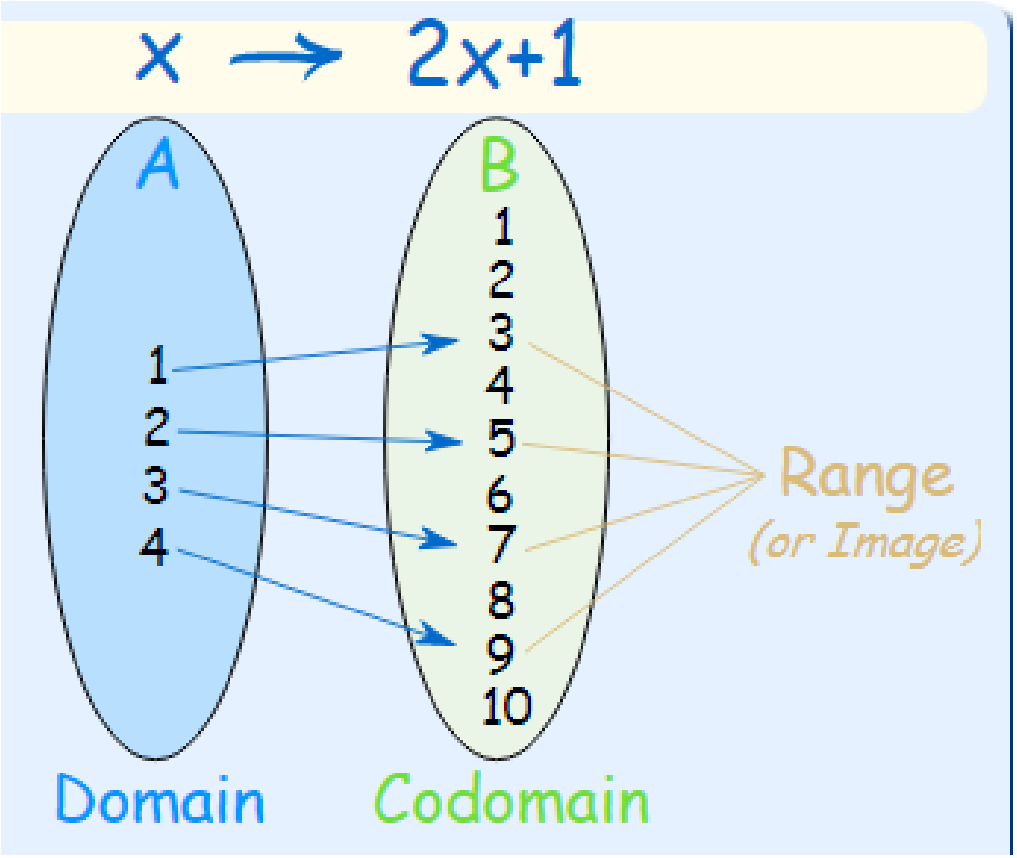
❸Ways to determine the output.

For example, $f: \mathbf{Z} \rightarrow \mathbf{R}$, $f(m) = 1/(m^2 + 3)$

2.3- Functions/Mapping/ Transformation - Ánh Xạ

codomain vs **range**

- Codomain contains outputs
- Range: the set of all outputs
- Range is a subset Codomain (maybe equal)



Functions as sets of ordered pairs

Set of Ordered Pairs

A function can then be defined as a **set** of ordered pairs:

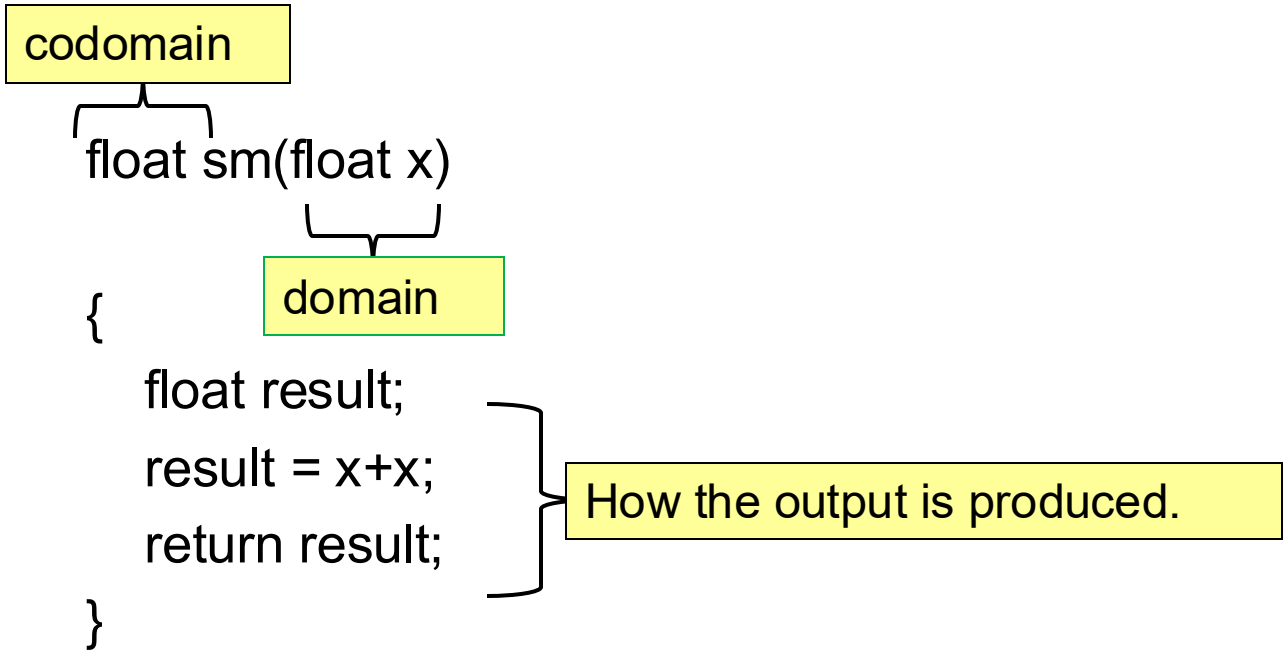
Example: **{(2,4), (3,5), (7,3)}** is a function that says

"2 is related to 4", "3 is related to 5" and "7 is related 3".

Also, notice that:

- the domain is **{2,3,7}** (the input values)
- and the range is **{4,5,3}** (the output values)

A function in programming language (C)



Functions/Mappings/Transformations…

What are functions?

1. $f: ? \rightarrow ? : f(x) = x^2 + 2$
2. $f: ? \rightarrow ? : f(x) = 1/(x-1)^2 + 5x$
3. $f: ? \rightarrow ? : f(x) = (2x+5)/7$
4. $f: ? \rightarrow ? : f(x) = (2x+5)^2/(7-2x)$

Some Important Functions

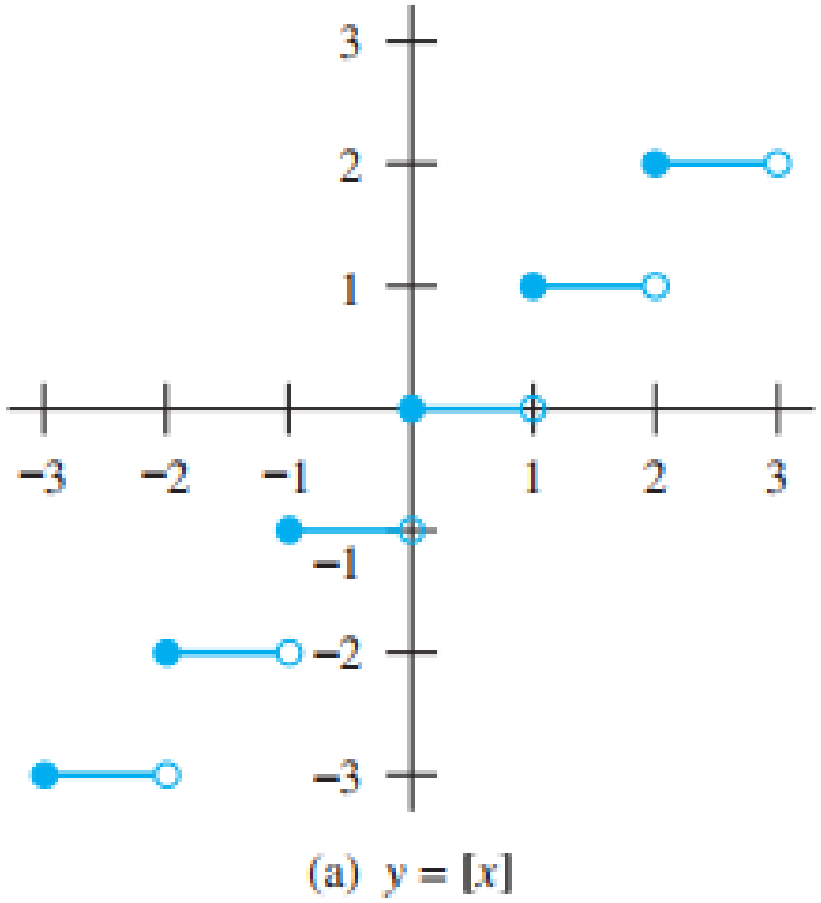
Floor function

f: ? → ? such that f(x)= ⌊x⌋ = largest integer that less than or equal to x (số nguyên lớn nhất chưa vượt qua x), ⌊x⌋ ?x.

EX:

$$\left\lfloor \frac{1}{2} \right\rfloor = 0, \left\lfloor -\frac{1}{2} \right\rfloor = -1,$$

$$\lfloor 3.1 \rfloor = 3, \lfloor 7 \rfloor = 7$$



(a) $y = [x]$

Some Important Functions

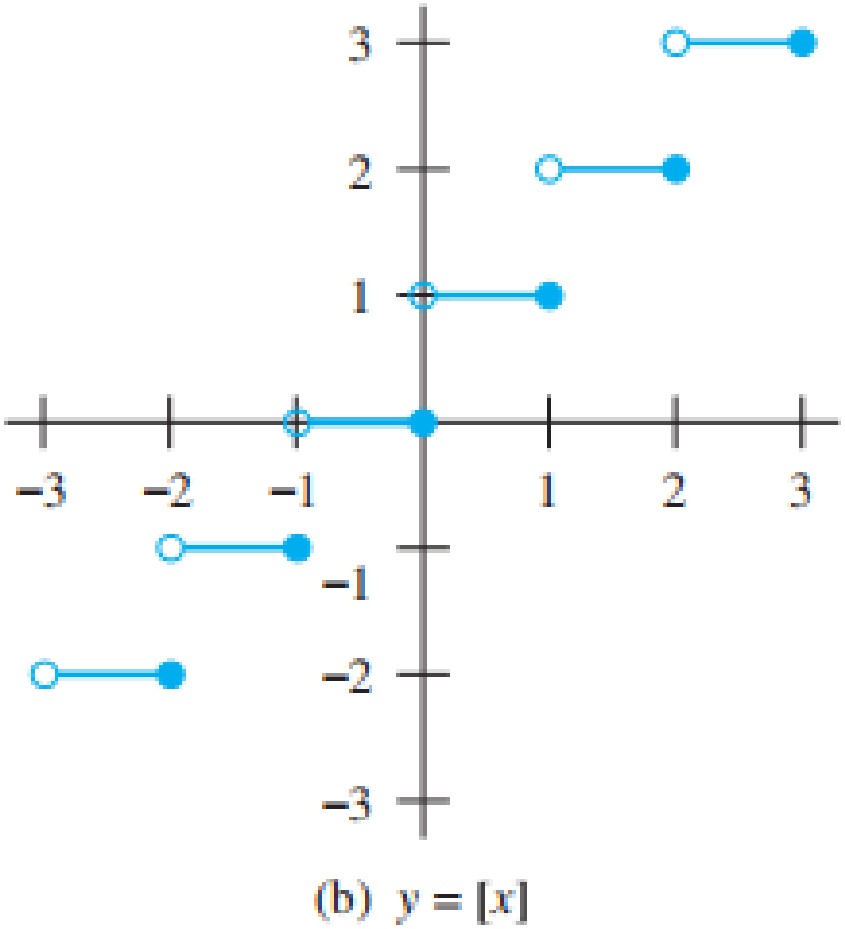
Ceiling function

f: ℝ → ℤ such that f(x)= $\lceil x \rceil$ = smallest integer that greater than or equal to x (số nguyên bé nhất x chưa vượt qua), x ∈ $\lceil x \rceil$.

Ex:

$$\left\lceil \frac{1}{2} \right\rceil = 1, \left\lceil -\frac{1}{2} \right\rceil = 0,$$

$$\lceil 3.1 \rceil = 4, \lceil 7 \rceil = 7$$

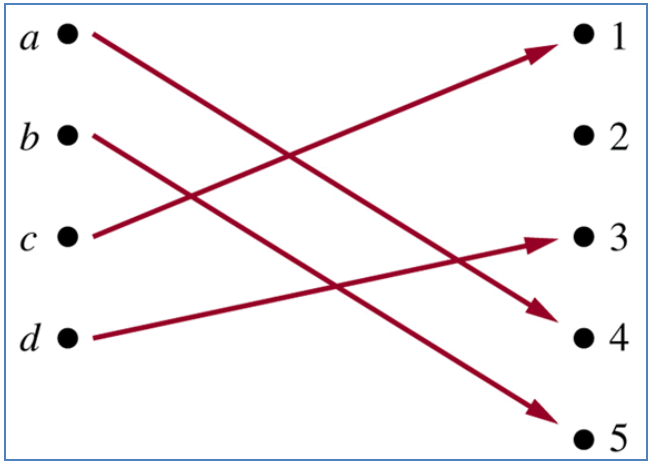


(b) $y = \lceil x \rceil$

One-to-One/ Injective functions (đơn ánh)

Function f is one-to-one (or injective) if and only if **a ≠ b → f(a) ≠ f(b)**

for all a and b in the domain of f.



- $f(x) = x^3$

Different inputs: $x \neq y \rightarrow x^3 \neq y^3$: different outputs

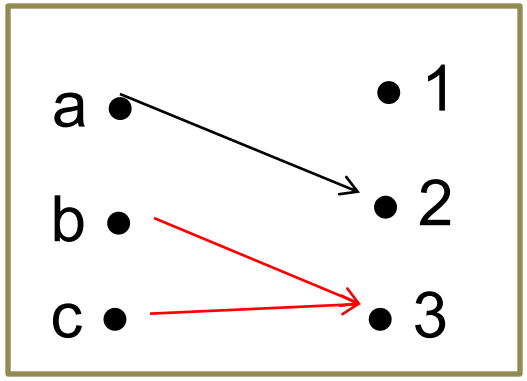
$$\forall x \forall y(x \neq y \rightarrow f(x) \neq f(y))$$

- Not one-to-one:

There are two different inputs that map to the same output

$$\exists x \exists y(x \neq y \wedge f(x) = f(y))$$

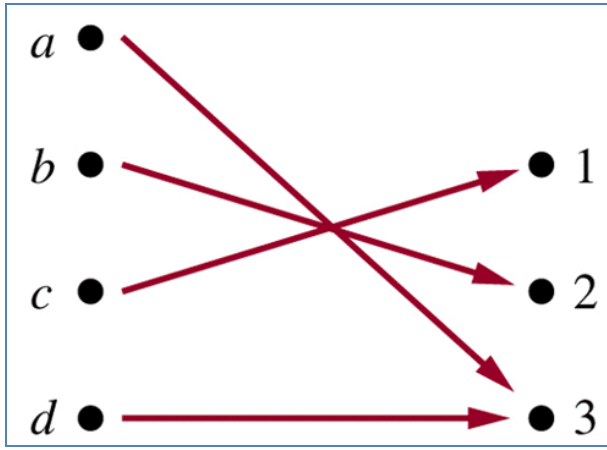
$$f(b) = f(c) = 3$$



Not one-to-one

Onto Functions – Ánh xạ trên (toàn ánh)

A function f from A to B is called **onto, or surjective**, iff for every element b in B there is an element a in A with f(a)=b.



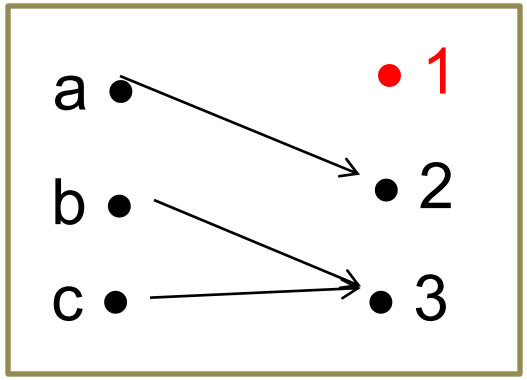
- f: ℤ → ℤ, f(m) =m-1

  f is **onto** because ∀y∈ ℤ, y=f(m)=m-1, where m=y+1

Not onto:

$$\exists y \in B \forall x \in A(f(x) \neq y)$$

f(a) ≠ 1, f(b) ≠ 1, f(c) ≠ 1

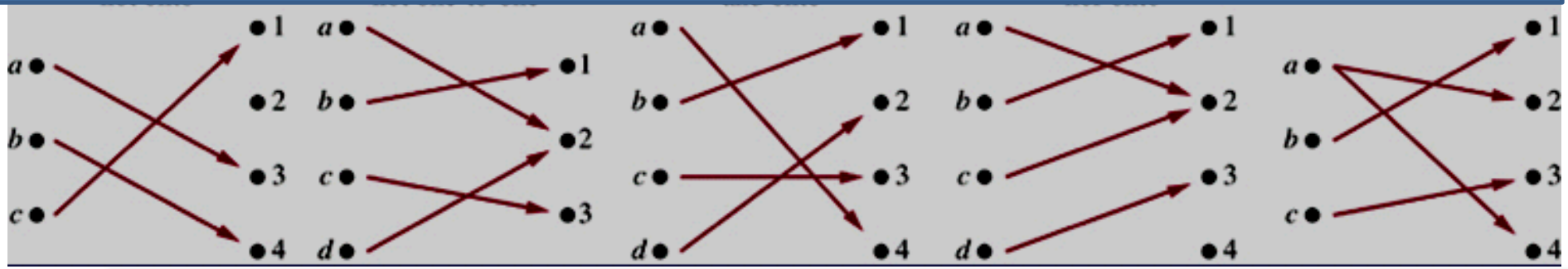


Not onto

One-to-one Correspondent / Bijective Functions (song ánh)

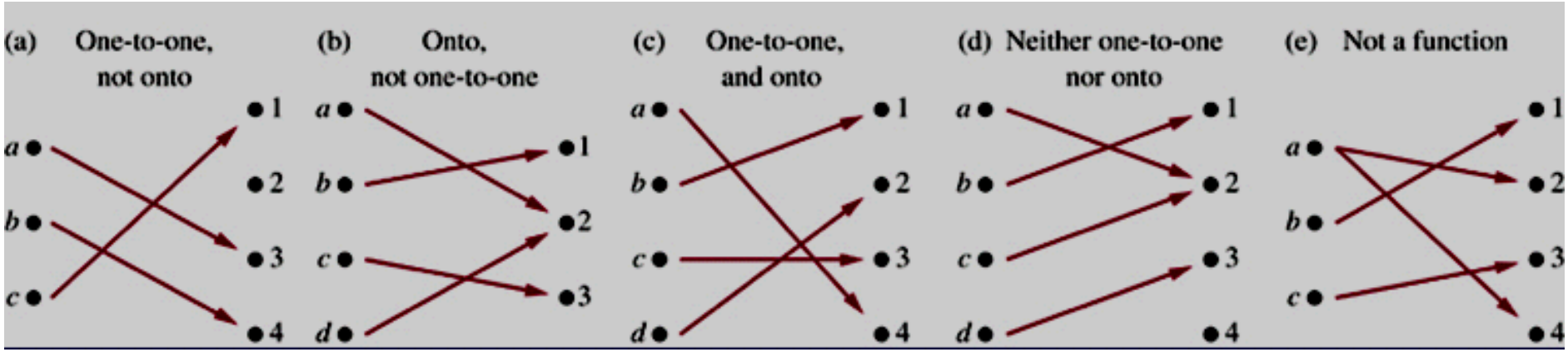
Function f is a **one-to-one corespondence** or a **bijection** if it is both one-to-one and onto.

f: {A,B,…,Z} →{65,66,…,90} is a bijection



One-to-one Correspodent / Bijective Functions (song ánh)

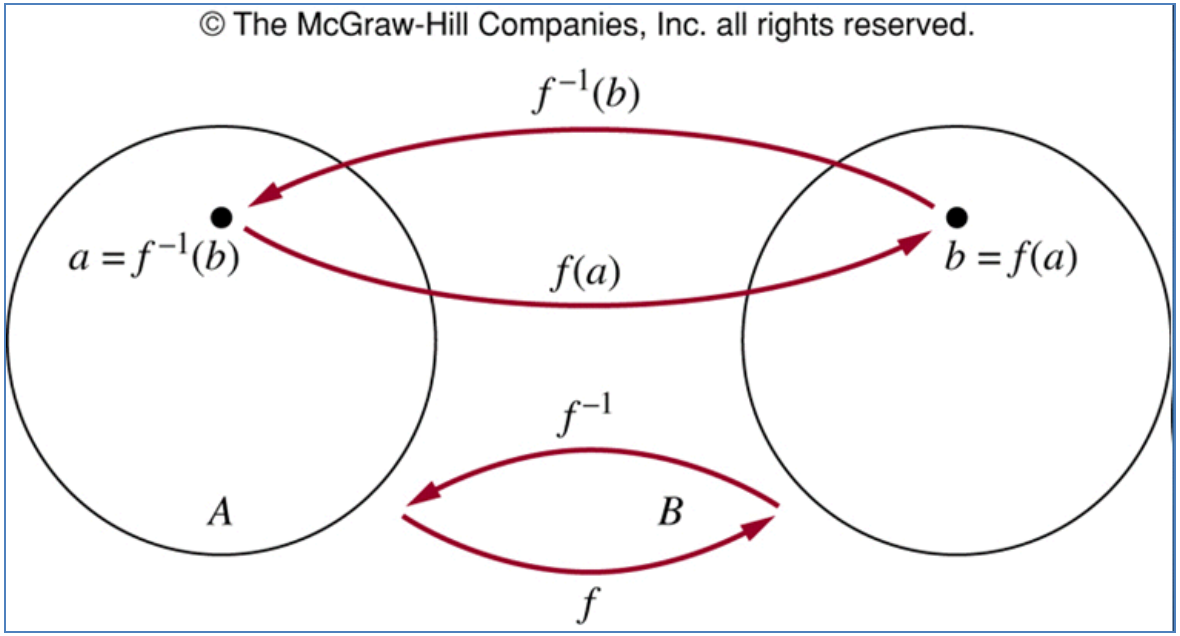
Function f is a **one-to-one corespondence** or a **bijection** if it is both one-to-one and onto.



Inverse Functions

Let f is a bijection from A to B. The *inverse function*, denoted by f^{-1}, of f is the function that assigns to an element b belonging to B the unique element a in A such that f(a)=b. Hence $f^{-1}(b)=a$ when $f(a)=b$.

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Inverse functions.

Examples.

1. $f = \{(A,10), (B,11), (C,12), (D,13), (E,14), (F,15)\}$

➔$f^{-1} = \{(10,A), (11,B), (12,C), (13,D), (14,E), (15,F)\}$

2. $f(x) = 3x + 5 = y$ ➔ $x = (y-5)/3$

➔$f^{-1}(y) = x = (y-5)/3$

Inverse Functions…

f:?→? such that f(x)=x+1

Is f invertible? And if it is, what is its inverse?

Step 1: Show that f is onto

f(y-1)=y for all y

➔ f is **onto**

Step 2: Show that f is one-to-one

f(a)=a+1= f(b)=b+1 ➔ a=b ➔ f is **one-to-one**

➔ f is **bijection** ➔ f is **invertible**

Step 3: Find **inverse function**

f(x)= y=x+1 ➔$x=f^{-1}(y)$

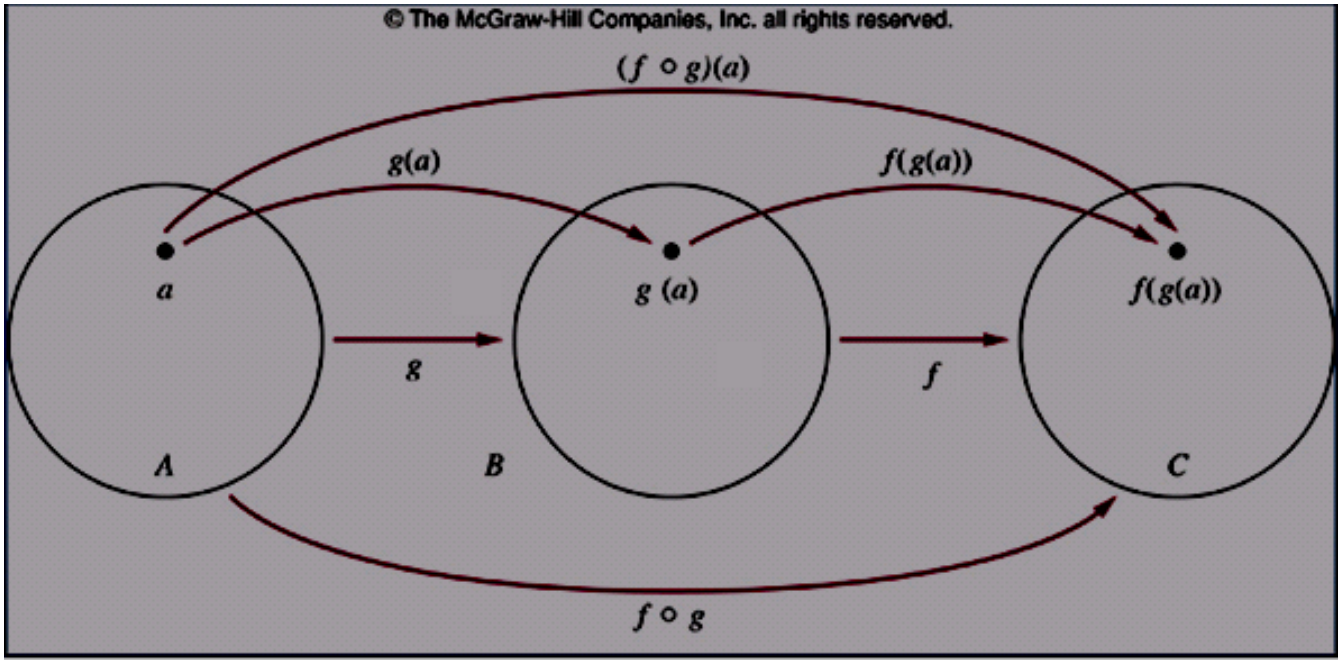
x=y-1 ➔ $f^{-1}(y)=y-1$

Composition of Functions – Ánh xạ hợp

Let g:A → B, f: B → C

The *composition* of f and g, denoted by f∘g, is defined by:

(f∘g)(x)= f(g(x))



Example:

f: ℝ →ℝ, f(x)=x+1

g:ℝ→ℝ, g(x)= x^2

(f∘g)(x)= f(g(x))= f(x^2) = x^2+1

(g∘f)(x) = g(f(x)) = g(x+1)= $(x+1)^2$

Exercises

8. Find these values.

a) $\lfloor 1.1 \rfloor$
b) $\lceil 1.1 \rceil$
c) $\lfloor -0.1 \rfloor$
d) $\lceil -0.1 \rceil$
e) $\lceil 2.99 \rceil$
f) $\lceil -2.99 \rceil$
g) $\lfloor \frac{1}{2} + \lceil \frac{1}{2} \rceil \rfloor$
h) $\lceil \lfloor \frac{1}{2} \rfloor + \lceil \frac{1}{2} \rceil + \frac{1}{2} \rceil$

9. Find these values.

a) $\lceil \frac{3}{4} \rceil$
b) $\lfloor \frac{7}{8} \rfloor$
c) $\lceil -\frac{3}{4} \rceil$
d) $\lfloor -\frac{7}{8} \rfloor$
e) $\lceil 3 \rceil$
f) $\lfloor -1 \rfloor$
g) $\lfloor \frac{1}{2} + \lceil \frac{3}{2} \rceil \rfloor$
h) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor$

Exercises

10. Determine whether each of these functions from {*a, b, c, d*} to itself is one-to-one.

a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$

b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

11. Which functions in Exercise 10 are onto?

22. Determine whether each of these functions is a bijection from **R** to **R**.

a) $f(x) = -3x + 4$

b) $f(x) = -3x^2 + 7$

c) $f(x) = (x + 1)/(x + 2)$

d) $f(x) = x^5 + 1$

Exercises

3. Determine whether *f* is a function from the set of all bit strings to the set of integers if

a) *f (S)* is the position of a 0 bit in *S*.

b) *f (S)* is the number of 1 bits in *S*.

c) *f (S)* is the smallest integer *i* such that the *i*th bit of *S* is 1 and *f (S)* = 0 when *S* is the empty string, the string with no bits.

Exercises

20. Give an example of a function from **N** to **N** that is

a) one-to-one but not onto.

b) onto but not one-to-one.

c) both onto and one-to-one (but different from the identity function).

d) neither one-to-one nor onto.

21. Give an explicit formula for a function from the set of integers to the set of positive integers that is

a) one-to-one, but not onto.

b) onto, but not one-to-one.

c) one-to-one and onto.

d) neither one-to-one nor onto.

2.4- Sequences

- Sequence : $a_1, a_2, a_3, \ldots, a_n, \ldots$

  Ex: 1,3,5,8 : Finite sequence

  Ex: 1, 1, 2, 3, 5, 8, 13,… : Infinite sequence
- A sequence is a function from a subset of integers to a set S.
- a_n : image of the integer n
- a_i : a term of the sequence
- $\{a_n = 1/n\}$: $\mathbb{Z}^+ \rightarrow \mathbb{R}$ ➔ 1, 1/2, 1/3, 1/4, …

Sequence & sums

- Arithmetic progression: (cấp số cộng)

a, a + d, a + 2d, a + 3d, …, a + nd, …

d: common difference

3, 7, 11, 15, 19, 23, …, 3 + 4n, …

➔ $a_n = 3 + 4n$

- Geometric progression: (cấp số nhân)

a, ar, ar^2, ar^3, …, ar^n, …

r : common ratio

2, 1, ½, ¼, 1/8, 1/16, …, $2.(1/2)^n$, …

➔ $b_n = 2(1/2)^n$

Some Useful Sequences

nth Term	*First 10 Terms*
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, . . .
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, . . .
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, . . .
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, . . .
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, . . .
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, . . .

Hints for deducing a possible formula for the terms of a sequence.

1. 1 2 2 3 3 3 4 4 4 4……
2. 1 5 9 13 17 … 2 6 18 54 ….
3. 1 1 2 3 5 8 13 …
4. $\{a_n\}$ 1 7 25 79 241 727 ….

Hints for deducing a possible formula for the terms of a sequence.

1. Are there runs of same value? 1 2 2 3 3 3 4 4 4 4……
2. Are terms obtained from previous term by adding/ multiplying by a particular amount?

   1 5 9 13 17 … 2 6 18 54 ….
3. Are terms obtained by combining previous terms in a certain way? 1 1 2 3 5 8 13 …
4. Are they cycles among terms

Ex:

$\{a_n\}$ 1 7 25 79 241 727 ….

6 18 54 162 486 ➔ close to 3 ➔ $\{3^n\}$

$\{3^n\}$ = 3 9 27 81 243 729 …

➔ $\{a_n\} = \{3^n - 2\}$

2.5- Summations

$$a_m + a_{m+1} + a_{m+2} + \ldots + a_n = \sum_{j=m}^{n} a_j = \sum_{j=m}^{n} a_j = \sum_{m \le j \le n} a_j$$

a : Sequence

j : Index of summation

m: Lower limit

n : Upper limit

2.5- Summations

Examples :

What is the value of $\sum_{j=1}^{5} j^2$?

What is the value of $\sum_{k=4}^{8} (-1)^k$?

Find $\sum_{i=1}^{4}\sum_{j=1}^{3} ij$

Summations….

Theorem 1- (Summation of geometric series)

If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^{n} ar^j = \begin{cases} \dfrac{ar^{n+1} - a}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1. \end{cases}$$

Some Useful Summation Formulae

Sum	Closed Form
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x \rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty}, kx^{k-1}, \lvert x \rvert < 1$	$\frac{1}{(1-x)^2}$

Some Useful Summation Formulae

Examples : Find $\sum_{k=50}^{100} k^2$.

Solution: First note that because $\sum_{k=1}^{100} k^2 = \sum_{k=1}^{49} k^2 + \sum_{k=50}^{100} k^2$, we have

$$\sum_{k=50}^{100} k^2 = \sum_{k=1}^{100} k^2 - \sum_{k=1}^{49} k^2.$$

Using the formula $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$ we see that

$$\sum_{k=50}^{100} k^2 = \frac{100 \cdot 101 \cdot 201}{6} - \frac{49 \cdot 50 \cdot 99}{6} = 338{,}350 - 40{,}425 = 297{,}925.$$

Cardinality – Lực Lượng

- Cardinality = number of elements in a set.
- The sets A and B have the same cardinality if and only if there is a one-to-one correspondence from A to B
- A set that is either finite or has the same cardinality as the set of positive integers is called ***countable***.
- A set that is not countable is called ***uncountable.***
- When a infinite set S is countable , we denote the cardinality of S is $|S| = \aleph_0$ (aleph null)
- For example, $|\square| = \aleph_0$ because $\square$ is countable and infinite but $\square$ is uncountable and infinite, and we say $|\square| = 2^{\aleph_0}$

Examples p.159, 160

sets	countable	uncountable	cardinality
{a, b, …, z}, {x| $x^5 - 3x^2 - 11 = 0$}, …	✓		< ?
{0, 2, 4, …, }	✓		$\aleph_0$
N, Z^+, Z, Q, Z?Z, …	✓		$\aleph_0$
{x| 0 < x < 1}, R,…		✓	$2^{\aleph_0}$

Summary

- Sets
- Set operations
- Functions
- Sequences
- Summations